THE UNIVERSAL FIRST ORDER THICKENING OF AN IMMERSION OF ALGEBRAIC SPACES IS THE COMPLEMENT OF THE INFINITESIMAL NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3152-3166 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3168-3182, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Immersions of algebraic spaces as representable morphisms, universal first order thickenings, and conormal sheaves; the reduction is retained.

Source reference. Original lemma definition-conormal-sheaf: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 410–415. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma definition-first-order-infinitesimal-neighbourhood: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2325–2332. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma definition-universal-thickening: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3128–3138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-first-order-infinitesimal-neighbourhood: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2341–2368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-infinitesimal-neighbourhood-conormal: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2395–2409. Cited by the retained statement or proof but not part of the collected spans;

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supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-unramified-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2829–2837. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-closed-immersion-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3182–3182. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-open-immersion-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3182–3182. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-lemma-representable-transformations-property-implication: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3182–3182. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to X$ be an immersion of algebraic spaces over S. Then*

(1) *i is formally unramified,*
(2) *the universal first order thickening of Z over X is the first order infinitesimal neighbourhood of Z in X of Definition definition-first-order-infinitesimal-neighbourhood,*
(3) *the conormal sheaf of i in the sense of Definition definition-conormal-sheaf agrees with the conormal sheaf of i in the sense of Definition definition-universal-thickening.*

Proof. An immersion of algebraic spaces is by definition a representable morphism. Hence by Morphisms, Lemmas morphisms-lemma-open-immersion-unramified and morphisms-lemma-closed-immersion-unramified an immersion is unramified (via the abstract principle of Spaces, Lemma spaces-lemma-representable-transformations-property-implication). Hence it is formally unramified by Lemma lemma-unramified-formally-unramified. The other assertions follow by combining Lemmas lemma-first-order-infinitesimal-neighbourhood and lemma-infinitesimal-neighbourhood-conormal and the definitions. □